Module 1: Foundations and Limit Superior/Inferior

Proof language, sets, countability, sequences, and indicator functions

1. Mathematical Statements and Proof Language

Quantifiers. The symbols $\forall$ (for all) and $\exists$ (there exists) specify the logical scope of a statement. Their order matters:

$$\forall x \in \mathbb{R}\, \exists y \in \mathbb{R} :\ y > x \quad \text{is true,} \qquad \exists y \in \mathbb{R}\, \forall x \in \mathbb{R} :\ y > x \quad \text{is false.}$$

Implications. $P \Rightarrow Q$ means that P is *sufficient* for Q, while Q is *necessary* for P; $P \Leftrightarrow Q$ means both directions hold.

Negation of quantified statements. Quantifiers switch under negation:

$$\neg(\forall x\, P(x)) \iff \exists x\, \neg P(x), \qquad \neg(\exists x\, P(x)) \iff \forall x\, \neg P(x).$$

In particular, the distinction between "eventually" and "arbitrarily far along" is encoded by

$$\underbrace{\exists N\, \forall n \geq N :\ P(n)}_{\text{eventually always}} \qquad \text{versus} \qquad \underbrace{\forall N\, \exists n \geq N :\ P(n)}_{\text{infinitely often}}.$$

Common proof strategies include direct proof, contrapositive ($P \Rightarrow Q \iff \neg Q \Rightarrow \neg P$), contradiction, and induction.

2. Sets, Images, and Inverse Images

For $A, B \subseteq \Omega$, union and intersection correspond to logical OR and AND:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}, \qquad A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

The complement is $A^c = \Omega \setminus A$, the difference is $A \setminus B = A \cap B^c$, and the Cartesian product is $A \times B = \{(a, b) : a \in A, b \in B\}$. De Morgan's laws extend to any indexed family:

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \qquad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

For a map $f : X \to Y$, define the *direct image* and *inverse image* by

$$f[A] = \{f(x) : x \in A\} \quad (A \subseteq X), \qquad f^{-1}[B] = \{x \in X : f(x) \in B\} \quad (B \subseteq Y).$$

The preimage $f^{-1}[B]$ exists even if f has no inverse function. Crucially, preimages preserve *all* unions, intersections, and complements:

$$\boxed{\begin{aligned} f^{-1}\left[\bigcup_i B_i\right] &= \bigcup_i f^{-1}[B_i], \\ f^{-1}\left[\bigcap_i B_i\right] &= \bigcap_i f^{-1}[B_i], \\ f^{-1}[Y \setminus B] &= X \setminus f^{-1}[B]. \end{aligned}}$$

Direct images preserve unions, but generally not intersections or complements. For example, if $f(1) = f(2) = 0$, then $f[\{1\} \cap \{2\}] = \varnothing$ whereas $f[\{1\}] \cap f[\{2\}] = \{0\}$.

Key point. Inverse images respect the logical structure of subsets. This is why they are fundamental in the definitions of continuity and measurability.

3. Countability

A set is *countably infinite* if it admits a bijection with $\mathbb{N}$; *countable* means finite or countably infinite. Use $\mathbb{N} = \{1, 2, 3, \ldots\}$ below.

$\mathbb{Z}$ is countable. One enumeration is

$$0, 1, -1, 2, -2, 3, -3, \ldots,$$

which lists every integer exactly once, hence defines a bijection $\mathbb{N} \leftrightarrow \mathbb{Z}$.

$\mathbb{Q}$ is countable. Every rational number has a representation p/q with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. The pairs (p, q) can be enumerated, for example, by increasing $|p| + q$ (only finitely many pairs occur at each level). Enumerate the corresponding fractions and skip repetitions; every rational eventually appears. Since $\mathbb{Q}$ is infinite, this gives a bijection $\mathbb{N} \leftrightarrow \mathbb{Q}$.

$\mathbb{R}$ is uncountable (Cantor diagonal argument). It suffices to show that $(0, 1)$ cannot be enumerated. Suppose, toward a contradiction, that

$$(0, 1) = \{x_1, x_2, \ldots\}, \qquad x_n = 0.a_{n1}a_{n2}a_{n3}\ldots$$

where each decimal uses the expansion not ending in an infinite string of 9's. Construct $y = 0.b_1b_2\ldots$ by

$$b_n = \begin{cases} 1, & a_{nn} \neq 1, \\ 2, & a_{nn} = 1. \end{cases}$$

Then $y \in (0, 1)$ and its nth digit differs from that of x_n, so $y \neq x_n$ for every n. This contradicts the assumed complete enumeration.

Key point. For $\mathbb{Z}$ and $\mathbb{Q}$, *one* bijection proves countability. For $(0, 1)$, diagonalization defeats *every* proposed enumeration.

4. Sequences and Their Limit Superior/Inferior

4.1 Convergence and bounds (brief review)

A real sequence converges to L if $\forall \varepsilon > 0\, \exists N\, \forall n \geq N$, $|x_n - L| < \varepsilon$. For a nonempty bounded set $S \subseteq \mathbb{R}$, $\sup S$ is its least upper bound and $\inf S$ its greatest lower bound; neither need belong to S. The least-upper-bound property of $\mathbb{R}$ is a formulation of completeness.

4.2 Numerical sequences

For $(x_n) \subseteq \mathbb{R}$, define the upper and lower bounds of the Nth tail by

$$s_N = \sup_{n \geq N} x_n, \qquad t_N = \inf_{n \geq N} x_n.$$

Deleting initial terms can only decrease s_N and increase t_N:

$$s_{N+1} \leq s_N, \qquad t_{N+1} \geq t_N.$$

Thus both auxiliary sequences have limits in the *extended reals* $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, and we define

$$\boxed{\begin{aligned} \limsup_{n\to\infty} x_n &= \lim_{N\to\infty} s_N = \inf_N \sup_{n \geq N} x_n, \\ \liminf_{n\to\infty} x_n &= \lim_{N\to\infty} t_N = \sup_N \inf_{n \geq N} x_n. \end{aligned}}$$

The outer inf and sup arise because the tail upper bounds decrease and the tail lower bounds increase. They are *asymptotic* bounds, not necessarily the bounds of the whole sequence.

For a bounded sequence, $\limsup x_n$ and $\liminf x_n$ are respectively the largest and smallest *subsequential limits*. The limit values need not be attained by any term. In addition,

$$\liminf x_n \leq \limsup x_n, \qquad x_n \to L \in \mathbb{R} \iff \liminf x_n = \limsup x_n = L.$$

For instance, $x_n = (-1)^n + 1/n$ satisfies $\limsup x_n = 1$ and $\liminf x_n = -1$; an isolated first term of 100 would not affect either limit. An unbounded sequence may have an infinite upper or lower limit (e.g., $x_n = (-1)^n n$).

4.3 Sequences of sets

Let $A_n \subseteq \Omega$. In the inclusion order on $\mathcal{P}(\Omega)$, the supremum of a family is its union and the infimum is its intersection. Define the tail upper and lower bounds

$$B_N = \sup_{n\geq N} A_n = \bigcup_{n\geq N} A_n, \qquad C_N = \inf_{n\geq N} A_n = \bigcap_{n\geq N} A_n.$$

As N increases, B_N decreases under inclusion while C_N increases. Consequently,

$$\boxed{\begin{aligned} \limsup_{n\to\infty} A_n &= \inf_N B_N = \bigcap_{N=1}^{\infty} \bigcup_{n\geq N} A_n, \\ \liminf_{n\to\infty} A_n &= \sup_N C_N = \bigcup_{N=1}^{\infty} \bigcap_{n\geq N} A_n. \end{aligned}}$$

Equivalently, expanding the membership quantifiers,

$$\begin{aligned} x \in \limsup A_n &\iff \forall N\, \exists n \geq N :\ x \in A_n \quad \text{(infinitely often)}, \\ x \in \liminf A_n &\iff \exists N\, \forall n \geq N :\ x \in A_n \quad \text{(eventually always)}. \end{aligned}$$

These descriptions are *consequences* of the order-theoretic definitions: the outer operation combines all tails, not just one fixed tail.

5. The Order-Theoretic Bridge: Indicator Functions

The power set $(\mathcal{P}(\Omega), \subseteq)$ is a *complete lattice*: every family of subsets has a supremum (union) and an infimum (intersection). It is therefore meaningful to use the same tail sup/inf construction for numbers and sets.

For $A \subseteq \Omega$, its *indicator function* is

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases} \qquad \text{Thus} \qquad A = \{x : \mathbf{1}_A(x) = 1\}.$$

It translates set membership into a numerical value, without losing information. In fact $A \mapsto \mathbf{1}_A$ is an *order isomorphism* from $(\mathcal{P}(\Omega), \subseteq)$ to the $\{0,1\}$-valued functions on Ω with pointwise order:

$$\begin{gathered} A \subseteq B \iff \mathbf{1}_A \leq \mathbf{1}_B, \qquad \mathbf{1}_{A^c} = 1 - \mathbf{1}_A, \qquad \mathbf{1}_{A\cap B} = \mathbf{1}_A\, \mathbf{1}_B, \\ \mathbf{1}_{\bigcup_i A_i}(x) = \sup_i \mathbf{1}_{A_i}(x), \qquad \mathbf{1}_{\bigcap_i A_i}(x) = \inf_i \mathbf{1}_{A_i}(x). \end{gathered}$$

Apply these identities successively to the two layers of the set limit:

$$\begin{aligned} \mathbf{1}_{\limsup A_n}(x) &= \mathbf{1}_{\bigcap_N \bigcup_{n\geq N} A_n}(x) = \inf_N \sup_{n\geq N} \mathbf{1}_{A_n}(x) = \limsup_n \mathbf{1}_{A_n}(x), \\ \mathbf{1}_{\liminf A_n}(x) &= \sup_N \inf_{n\geq N} \mathbf{1}_{A_n}(x) = \liminf_n \mathbf{1}_{A_n}(x). \end{aligned}$$

For a fixed x, the sequence $\mathbf{1}_{A_n}(x)$ takes values only in $\{0,1\}$. Its limsup equals 1 exactly when x appears infinitely often, while its liminf equals 1 exactly when x eventually appears at every index.

Moreover, $\liminf A_n \subseteq \limsup A_n$: for any m, N, choosing $k = \max\{m, N\}$ gives

$$C_m \subseteq A_k \subseteq B_N \quad \Longrightarrow \quad \sup_m C_m \subseteq \inf_N B_N.$$

Finally, convergence of sets in this sense is exactly pointwise convergence of indicators:

$$\boxed{\liminf A_n = \limsup A_n = A \iff \mathbf{1}_{A_n}(x) \to \mathbf{1}_A(x) \quad \forall x \in \Omega.}$$

Key point. The indicator map turns a problem about sets into one about numerical functions. Later, measure and probability will use the same bridge: $\mu(A) = \int \mathbf{1}_A\, d\mu$ and $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}_A]$.